

The core of adjoint functors

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ABSTRACT. There is a lot of redundancy in the usual definition of adjoint functors. We define and prove the core of what is required.

Kan [1] introduced the notion of adjoint functors. By defining the unit and counit natural transformations, he paved the way for the notion to be internalized to any 2-category. This was done by Kelly [2] whose interest at the time was particularly in the 2-category of $\mathcal{V}$ -categories for a monoidal category $\mathcal{V}$ (in the sense of Eilenberg-Kelly [3]).

For $\mathcal{V}$ -categories $\mathcal{A}$ and $\mathcal{X}$, an *adjunction* consists of

- (1) $\mathcal{V}$ -functors $U : \mathcal{A} \rightarrow \mathcal{X}$ and $F : \mathcal{X} \rightarrow \mathcal{A}$;
- (2) a $\mathcal{V}$ -natural family of isomorphisms $\pi : \mathcal{A}(F X, A) \cong \mathcal{X}(X, U A)$ in $\mathcal{V}$ indexed by $A \in \mathcal{A}$, $X \in \mathcal{X}$.

We write $\pi : F \dashv U : \mathcal{A} \rightarrow \mathcal{X}$.

The following result is well known; for example see Section 1.11 of [4].

Proposition 0.1. *Suppose $U : \mathcal{A} \rightarrow \mathcal{X}$ is a $\mathcal{V}$ -functor, $F : ob \mathcal{X} \rightarrow ob \mathcal{A}$ is a function, and, for each $X \in \mathcal{X}$, $\pi : \mathcal{A}(F X, A) \cong \mathcal{X}(X, U A)$ is a family of isomorphisms $\mathcal{V}$ -natural in $A \in \mathcal{A}$. Then there exists a unique adjunction $\pi : F \dashv U : \mathcal{A} \rightarrow \mathcal{X}$ for which $F : ob \mathcal{X} \rightarrow ob \mathcal{A}$ is the effect of the $\mathcal{V}$ -functor $F : \mathcal{X} \rightarrow \mathcal{A}$ on objects.*

Definition 0.2. For $\mathcal{V}$ -categories $\mathcal{A}$ and $\mathcal{X}$, an *adjunction core* consists of

- (1) functions $U : ob \mathcal{A} \rightarrow ob \mathcal{X}$ and $F : ob \mathcal{X} \rightarrow ob \mathcal{A}$;
- (2) a family of isomorphisms $\pi : \mathcal{A}(F X, A) \cong \mathcal{X}(X, U A)$ in $\mathcal{V}$ indexed by $A \in \mathcal{A}$, $X \in \mathcal{X}$.

Given such a core, we make the following definitions:

- (a) $\eta_X : X \rightarrow U F X$ is the composite

$$I \xrightarrow{j} \mathcal{A}(F X, F X) \xrightarrow{\pi} \mathcal{X}(X, U F X);$$

- (b) $\varepsilon_A : F U A \rightarrow A$ is the composite

$$I \xrightarrow{j} \mathcal{X}(U A, U A) \xrightarrow{\pi^{-1}} \mathcal{A}(F U A, X);$$

(c) $U_{AB} : \mathcal{A}(A, B) \longrightarrow \mathcal{X}(U A, U B)$ is the composite

$$\mathcal{A}(A, B) \xrightarrow{\mathcal{A}(\varepsilon_A, 1)} \mathcal{A}(F U A, B) \xrightarrow{\pi} \mathcal{X}(U A, U B);$$

(d) $F_{XY} : \mathcal{X}(X, Y) \longrightarrow \mathcal{A}(F X, F Y)$ is the composite

$$\mathcal{X}(X, Y) \xrightarrow{\mathcal{X}(1, \eta_Y)} \mathcal{X}(X, U F Y) \xrightarrow{\pi^{-1}} \mathcal{A}(F X, F Y).$$

Clearly each adjunction $\pi : F \dashv U : \mathcal{A} \longrightarrow \mathcal{X}$ includes an adjunction core as part of its data. Then it follows directly from the Yoneda lemma and the definitions (a) and (b) that the effect of U and F on homs are as in (c) and (d).

Theorem 0.3. *An adjunction core extends to an adjunction if and only if one of the diagrams (0.1) or (0.2) below commutes. The adjunction is unique when it exists.*

$$(0.1) \quad \begin{array}{ccc} \mathcal{A}(A, B) \otimes \mathcal{A}(F X, A) & \xrightarrow{U_{AB} \otimes \pi} & \mathcal{X}(U A, U B) \otimes \mathcal{X}(X, U A) \\ \text{comp} \downarrow & = & \downarrow \text{comp} \\ \mathcal{A}(F X, B) & \xrightarrow{\pi} & \mathcal{X}(X, U B) \end{array}$$

$$(0.2) \quad \begin{array}{ccc} \mathcal{X}(Y, U A) \otimes \mathcal{X}(X, Y) & \xrightarrow{\pi^{-1} \otimes F_{XY}} & \mathcal{A}(F Y, A) \otimes \mathcal{A}(F X, F Y) \\ \text{comp} \downarrow & = & \downarrow \text{comp} \\ \mathcal{X}(X, U A) & \xrightarrow{\pi^{-1}} & \mathcal{A}(F X, A) \end{array}$$

Proof We deal first with the version involving diagram (0.1). For an adjunction, (0.1) expresses the $\mathcal{V}$ -naturality of π in $A \in \mathcal{A}$. Conversely, given an adjunction core satisfying (0.1), we paste to the left of (0.1) with $X = U C$, the diagram

$$(0.3) \quad \begin{array}{ccc} \mathcal{A}(A, B) \otimes \mathcal{A}(C, A) & \xrightarrow{1 \otimes \mathcal{A}(\varepsilon_C, 1)} & \mathcal{A}(A, B) \otimes \mathcal{A}(F U C, A) \\ \text{comp} \downarrow & = & \downarrow \text{comp} \\ \mathcal{A}(C, B) & \xrightarrow{\mathcal{A}(\varepsilon_C, 1)} & \mathcal{A}(F U C, B) \end{array}$$

which commutes by naturality of composition. This leads to the following commutative square.

$$(0.4) \quad \begin{array}{ccc} \mathcal{A}(A, B) \otimes \mathcal{A}(C, A) & \xrightarrow{U \otimes U} & \mathcal{X}(U A, U B) \otimes \mathcal{X}(U C, U A) \\ \text{comp} \downarrow & = & \downarrow \text{comp} \\ \mathcal{A}(C, B) & \xrightarrow{U} & \mathcal{X}(U C, U B) \end{array}$$

We also have the equality

$$(0.5) \quad \left(I \xrightarrow{j} \mathcal{A}(A, A) \xrightarrow{U} \mathcal{X}(U A, U B) \right) = \left(I \xrightarrow{j} \mathcal{X}(U A, U B) \right)$$

straight from the definitions (b) and (c). Together (0.4) and (0.5) tell us that U is a $\mathcal{V}$ -functor. Now the general diagram (0.1) expresses the $\mathcal{V}$ -naturality of π in A . By Proposition 0.1, we have an adjunction determined uniquely by the core.

Writing $\mathcal{V}^{\text{rev}}$ for $\mathcal{V}$ with the reversed monoidal structure $A \otimes^{\text{rev}} B = B \otimes A$, and applying the first part of this proof to the $\mathcal{V}^{\text{rev}}$ -enriched adjunction $\pi^{-1} : U^{\text{op}} \dashv F^{\text{op}} : \mathcal{X}^{\text{op}} \longrightarrow \mathcal{A}^{\text{op}}$, which is the same as an adjunction $\pi : F \dashv U : \mathcal{A} \longrightarrow \mathcal{X}$, we see that it is equivalent to an adjunction core satisfying (0.2). $\square$

Corollary 0.4. *If $\mathcal{V}$ is a poset then adjunction cores are adjunctions.*

Proof All diagrams, including (0.1), commute in such a $\mathcal{V}$. $\square$

References

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